

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.8: Commercial Arithmetic 1 — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.8: Commercial Arithmetic 1
Driving Question	DRIVING QUESTION: How can Amina — and every Kenyan — use commercial arithmetic to make smart, fair, and informed money decisions in markets, clubs, and across borders?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Amina's KSh 4,500 — Anchoring the Phenomenon & Building a Club Budget	Learners examined the Environmental Club's trip budget for a one-day visit to Lake Victoria, sorting eight cost items (bus hire, entry fees, ugali-and-tilapia lunch, T-shirt printing, etc.) into fixed and variable categories, and calculated that the club's KSh 6,000 income falls KSh 6,250 short of the KSh 12,250 total expenditure needed.	A budget is a mathematical plan comparing income (money coming in) and expenditure (money going out). Costs can be fixed (they do not change with the number of participants, e.g., bus hire) or variable (they increase with each additional participant, e.g., entry fees). When total expenditure exceeds total income, there is a deficit — a shortfall that requires a mathematical solution.	Amina's confusion about her KSh 4,500 begins with the same problem the club faces: she has not yet written down all her expected costs and compared them to her income. Without a budget, she cannot know whether she truly has enough — or how much she is short — before she starts spending. This is why every Kenyan who wants to make a smart money decision, whether for a school trip, a smartphone purchase, or a cross-border transfer, must start with a budget.
Lesson 2 Discounts and Percentage Discounts — How Much Does Amina Really Save at the Kondele Electronics Stall?	Learners observed, through a role-play shopping scenario at a simulated Kondele electronics stall, that a '20% off' sign reduces the marked price of KSh 3,200 to a sale price of KSh 2,560 — a saving of KSh 640 — and that this saving is computed by multiplying the marked price by the percentage expressed as a fraction of 100, then subtracting.	Discount = Marked Price – Sale Price; Percentage Discount = (Discount ÷ Marked Price) × 100. These two formulas work in both directions: given a percentage discount you can find the sale price, and given both prices you can verify the percentage discount a trader claims to be offering.	Amina's '20% off' smartphone deal reduces the price from KSh 3,200 to KSh 2,560, which she can pay from her remaining KSh 3,700 — leaving KSh 1,140. Mathematics reveals that the advertised discount is real and makes the purchase affordable, though it leaves only KSh 1,140 for the Uganda transfer, showing why verifying discounts with arithmetic — rather than trusting a sign — is essential for smart money decisions.
Lesson 3 Discounts and	Learners worked through three Kenyan retail discount scenarios — a Naivas	Discount = Marked Price – Sale Price; Percentage Discount = (Discount ÷ Marked	Amina's instinct to chase the biggest percentage discount (25% at Naivas) would

Percentage Discounts (Part 2 / Practice) — Which Deal Gives Amina Better Value?	Weekend Special, a Gikomba mitumba stall, and a Kisumu school-uniform comparison — and discovered that a higher advertised percentage discount does not always result in a lower sale price. In the Amina comparison problem, the Kondele phone (20% off KSh 3,200 = KSh 2,560) was KSh 20 cheaper than the Naivas phone (25% off KSh 3,440 = KSh 2,580), despite having a smaller percentage discount.	Price) $\times 100$; Sale Price = Marked Price $\times (1 - \text{Percentage Discount} \div 100)$. To compare two discount deals fairly, you must calculate the actual sale price for each — you cannot simply compare percentage figures. A deal with a smaller percentage discount on a lower marked price may cost you less than a deal with a larger percentage discount on a higher marked price.	actually cost her KSh 20 more than buying at Kondele (20% off). This shows that the '20% off' advertisement on the Kondele stall is not misleading — it is genuinely the better deal for her, even though the percentage is smaller. For Amina to make a smart decision about her KSh 4,500, she must calculate the final sale price every time, not just compare the percentage figures on the advertisement boards.
Lesson 4 Commission and Percentage Commission — Who Gets Paid When Amina Sends Money to Uganda?	Learners observed — through a role-play at a simulated Kondele mobile-money kiosk — that an agent retains a fixed percentage of every transaction value as a commission. By counting dummy currency and recording data in a table, they saw that the commission amount grows proportionally with the transaction value, while the commission rate (percentage) stays constant at 3% across all three transactions.	Commission is an amount paid to an agent or intermediary for facilitating a financial transaction; it is calculated as $\text{Commission} = (\text{Commission Rate} \div 100) \times \text{Transaction Value}$. $\text{Percentage Commission} = (\text{Commission Amount} \div \text{Transaction Value}) \times 100$. Both formulas are mirrors of each other, just as Discount and Percentage Discount were mirrors in Lessons 2 and 3. The recipient of a money transfer always receives less than the amount sent, because the agent deducts a commission.	This explains a key part of Amina's phenomenon: when she sends money to her aunt in Uganda via a mobile-money agent, her aunt does not receive the full amount. If Amina sends KSh 1,140 and the agent charges 3% commission, the agent earns KSh 34.20 and only KSh 1,105.80 is left to convert into Uganda Shillings. For Amina to send enough money to cover her aunt's needs AND the agent's commission, she must calculate the gross amount to send — a reverse-commission problem that will be explored further in the sequence. The 'same KSh 4,500 producing different outcomes' in the driving question now has a new mathematical explanation: commissions are hidden costs that reduce the value of every transfer.
Lesson 5 Profit and Loss (Part 1 — Concepts & Calculation) — From Wakulima Market to Amina's Stall	Learners role-played as traders at Wakulima Market, buying and selling githeri, beans, second-hand clothes, and mandazi using price cards, and recorded cost prices and selling prices in a trading ledger. They observed that some transactions yielded profit while others resulted in a loss, and that the size of the margin varied widely between goods.	$\text{Profit} = \text{SP} - \text{CP}$ (when $\text{SP} > \text{CP}$); $\text{Loss} = \text{CP} - \text{SP}$ (when $\text{CP} > \text{SP}$); $\text{Percentage Profit} = (\text{Profit} \div \text{CP}) \times 100$; $\text{Percentage Loss} = (\text{Loss} \div \text{CP}) \times 100$. The percentage is always calculated relative to the cost price. The Kondele electronics stall would earn a 28% profit on the KSh 2,560 discounted smartphone if it paid KSh 2,000 for it, even after giving Amina a 20% discount.	Every price Amina encounters in the market — including the 'discounted' smartphone and the mandazi she sells — has a profit margin embedded in it. Understanding profit and loss mathematically allows Amina (and any Kenyan trader or consumer) to see beyond the surface price and evaluate whether a transaction is truly fair and financially sound.
Lesson 6 Profit and Loss (Part 2 — Application & Problem Solving) — From Nyama Choma to Amina's Mandazi Stall	Learners observed two multi-step scenarios — Mama Wanjiku's nyama choma stall in Kericho and Amina's mandazi stall — in which total cost price included both direct ingredient costs and overhead costs (charcoal, packaging). They observed that ignoring overhead costs produces an	Percentage profit and loss must always be calculated on the TOTAL cost price, which includes all overheads, not just the purchase price of goods. The multi-step framework is: (1) Total CP = direct costs + overheads; (2) Actual Revenue = SP $\times$ units actually sold; (3) Profit or Loss = Revenue –	Amina's KSh 4,500 is the profit she saved from ten weeks of mandazi sales (KSh 2,250 revenue a week); against weekly costs of KSh 1,800 for ingredients and fuel, her profit is KSh 450 a week (KSh 4,500 over the ten weeks). This means every shilling of her KSh 4,500 required careful

	inflated and misleading profit figure, and that in Amina's case her weekly revenue of KSh 2,250 was compared with her weekly costs of KSh 1,800.	Total CP; (4) % Profit or Loss = $(\text{Profit or Loss} \div \text{Total CP}) \times 100$. A desired selling price can be reverse-engineered from a target profit percentage using $\text{SP} = \text{CP} \times (1 + \% \text{Profit} \div 100)$.	cost management. When she now faces a discount at the Kondele stall, a commission on the mobile-money transfer, and a currency exchange for her aunt in Uganda, she is making decisions with money she earned through exactly the kind of commercial arithmetic this lesson practised — confirming that profit and loss calculations are foundational to all of Amina's financial decisions.
Lesson 7 Currency Conversion — Crossing Borders with Amina's Shillings	Learners examined a simulated CBK forex bureau exchange-rate table showing buying and selling rates for UGX, USD, and EUR, and observed that every currency has TWO different rates — a lower buying rate and a higher selling rate — and that using the wrong rate produces a significantly different (and incorrect) conversion result.	To convert KES into foreign currency, use the rate that matches the quote: for KES-per-unit rates (USD) use the selling rate and divide; for UGX-per-KES rates the bureau is buying your KES, so multiply by its buying rate. To convert back to KES, use the buying rate: multiply for USD, for UGX-per-KES rates divide by the selling rate. The difference between the buying and selling rate (the 'spread') is the forex bureau's embedded commission — a hidden profit that functions exactly like the percentage commission we studied in Lesson 4, but is concealed inside the rate rather than charged as a separate visible fee. Amina's KSh 1,140 less 3% commission (KSh 1,105.80), converted at the bureau's buying rate of 27.80 UGX per KES, yields approximately UGX 30,741 for her aunt in Kampala.	This explains part of Amina's original puzzle: the same KSh 4,500 produces different outcomes depending on which financial transaction she uses it for. When sending money to Uganda, the exchange rate she sees on the sign is NOT the rate she actually transacts at — the bureau uses its buying rate for her KES (which is less favourable to Amina) to make its own profit. Just as the mobile-money agent in Lesson 4 charged a visible commission, the forex bureau charges an invisible one through the rate spread. Understanding both forms of commission is essential for Amina — and every Kenyan — to make smart, fair, and informed cross-border money decisions.
Lesson 8 Synthesis, Model Revision & Final Explanation — Amina's Complete Financial Plan	Learners observed their own mathematical growth by comparing their rough Lesson 1 model of Amina's financial problem — drawn before any instruction — with their fully annotated Lesson 8 model, which correctly traces KSh 4,500 through budget allocation, a 20% discount on a KSh 3,200 smartphone (sale price KSh 2,560), a 3% mobile-money commission on the Uganda transfer, and a final currency conversion into Uganda Shillings at the bureau's buying rate of 27.80 UGX/KES. They also observed peers' financial plans and identified errors or omissions using the KICD rubric language.	Learners consolidated all six commercial arithmetic learning outcomes — budget preparation, discount and percentage discount, commission and percentage commission, percentage profit and loss, and currency conversion — into a single integrated financial plan. They learned that these operations are not independent: the amount available for currency conversion depends on the budget allocation, which in turn depends on the correct discount calculation and commission deduction. They also learned that smart financial decision-making requires checking each step before proceeding to the next, and that the direction of a currency transaction (KES	Amina's puzzle — how the same KSh 4,500 produces such different outcomes depending on discounts, commissions, exchange rates, and profit/loss — is now fully explained: each operation applies a specific mathematical formula to a specific base amount, and small percentage differences (e.g., using 28.50 instead of the bureau's buying rate of 27.80 for KES to UGX) produce real financial gains or losses. Amina — and any Kenyan — can make smart, fair, and informed money decisions by applying these six formulas correctly, in the right sequence, to the right base values, at every stage of a financial transaction.

		→ UGX vs. UGX → KES) determines which exchange rate applies.	
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